Name: _____

TOPIC TEST 2

Chapter 7 Logarithmic functions

- Time allowed: 70 minutes
- CAS calculators may be used.
- Part A: Five multiple-choice questions (5 marks).
- Part B: Four short-answer questions (9 marks).
- Part C: One analysis question (8 marks).

Part A

Multiple-choice questions

Five multiple-choice questions
1 mark each: 5 marks
Circle the correct answer.

1 Evaluate the logarithm $\log_7 126$ using the change of base formula. Round to 3 decimal places.

A $\frac{126}{7}$

B 2.485

C 0.402

D 2.890

E -2.890

[1 mark]

2 $\log\left(\frac{ab}{c}\right)$ is equal to:

A $[\log(a) \times \log(b)] \div \log(c)$

B $\log(a) + \log(b) - \log(c)$

C $ab \log\left(\frac{1}{c}\right)$

D $\left(\frac{1}{c}\right)\log_e(ab)$

E None of these

[1 mark]

3 If $f(x) = \log_e(2x)$, then $f'(1)$ equals:

A 1

B 2

C $\frac{1}{2}$

D $\log_e(2)$

E $2\log_e(2)$

[1 mark]

4 If $\log_{10}(7) = a$, then $\log_{10}\left(\frac{1}{70}\right)$

A $-(1 + a)$

B $\frac{a}{10}$

C $(1 + a)^{-1}$

D $1 + a$

E $\frac{1}{10a}$

[1 mark]

5 If $\log_x(y) = 100$ and $\log_2(x) = 10$, then the value of y is

A 2^1

B 2^{10}

C 2^{100}

D 2^{1000}

E $2^{10\,000}$

[1 mark]

Part B

Short-answer questions

Four short-answer questions
9 marks
Show your working where appropriate.

1 Find $\frac{dy}{dx}$ given $y = \log_e(2x + 1)$.

[1 marks]

2 Sketch the graph of the function $f(x) = -\ln(2x)$. Determine the range and the domain of the function.

[2 marks]

3 Anti-differentiate $\frac{6x}{x^2 + 1}$.

[2 marks]

4 Find the value of $\frac{1}{\log_3(60)} + \frac{1}{\log_3(60)} + \frac{1}{\log_3(60)}$.

[4 marks]

Part C

Analysis question

One analysis question
8 marks
Show your working where appropriate.

1 The annual growth rate for an investment that is growing continuously is given by $r = \frac{1}{t}\ln\left(\frac{A}{P}\right)$ where P is the principal and A is the amount after t years. An investment of \$10 000 in Dell Computer stock in 2009 grew to \$31 800 in 2012.

a Assuming the investment grew continuously, what was the annual growth rate (to 4 decimal places)?

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b If Dell continues to grow at the same rate, what will the $10 000 investment be worth in 2016?

c Assuming the investment grew continuously at the same rate, how long will it take for the $10 000 investment to grow to $500 000?

[8 marks]

[Total marks: 22]

Answers

Part A – Multiple-choice questions [1 mark each]

1 B

2 B

3 A

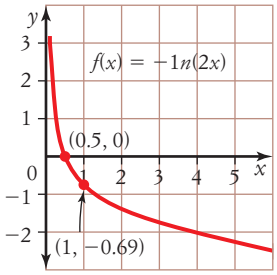
4 A

5 D [5 marks]

Part B – Short-answer questions

1 $\frac{dy}{dx} = \frac{2}{2x+1}$ [1 mark]

2



[2 marks]

Domain $(x : x > 0\}$, Range $\mathbf{R}$

3 Since $\int \frac{2x}{x^2+1}dx = \log_e(x^2+1) + c$ [1 mark]

then $\int \frac{6x}{x^2+1}dx = 3\int \frac{2x}{x^2+1}dx = 3\log_e(x^2+1) + c$ [1 mark]

4 $\frac{1}{\log_3(60)} + \frac{1}{\log_4(60)} + \frac{1}{\log_5(60)}$

$= \frac{1}{\frac{\log_{10}(60)}{\log_{10}(3)}} + \frac{1}{\frac{\log_{10}(60)}{\log_{10}(4)}} + \frac{1}{\frac{\log_{10}(60)}{\log_{10}(5)}}$ [1 mark]

$= \frac{\log_{10}(3)}{\log_{10}(60)} + \frac{\log_{10}(4)}{\log_{10}(60)} + \frac{\log_{10}(5)}{\log_{10}(60)}$ [1 mark]

$= \frac{\log_{10}(3) + \log_{10}(4) + \log_{10}(5)}{\log_{10}(60)}$

$= \frac{\log_{10}(3 \times 4 \times 5)}{\log_{10}(60)}$

$= \frac{\log_{10}(60)}{\log_{10}(60)}$ [1 mark]

$= 1$ [1 mark]

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Part C – Analysis questions

1 a $r = \frac{1}{3}\ln\left(\frac{31\ 800}{10\ 000}\right)$ **[1 mark]**

≈ 0.3856 **[1 mark]**

Therefore, the annual growth rate is 0.3856 or 38.56%.

b $0.3856 = \frac{1}{7}\ln\left(\frac{A}{10\ 000}\right)$ **[1 mark]**

$0.3856 \times 7 = \ln\left(\frac{A}{10\ 000}\right)$

$e^{(0.3856 \times 7)} = \frac{A}{10\ 000}$

$10\ 000e^{(0.3856 \times 7)} = A$ **[1 mark]**

$A = \$148\ 678.33$ **[1 mark]**

c $0.3856 = \frac{1}{t}\ln\left(\frac{50\ 000}{10\ 000}\right)$ **[1 mark]**

$0.3856 = \frac{1}{t}\ln(50)$

$0.3856t = \ln(50)$

$t = \frac{\ln(50)}{0.3856}$ **[1 mark]**

$t = 10.1453$ **[1 mark]**

Therefore, it will take 10.1453 years after the year 2009 for the $10 000 investment to grow to $500 000.

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